

Newman-Penrose Formalism Calculations

Stephani et al., Equation (31.38)

Metric

$$ds^2 = -2H(u, x, y) du \otimes du - du \otimes dv - W_x(u, x, y)^2 du \otimes dx + W_y(u, x, y)^2 du \otimes dy - dv \otimes du - W_x(u, x, y)^2 dx \otimes du + dx \otimes dx + W_y(u, x, y)^2 dy \otimes du + dy \otimes dy$$

Coordinates

$$(x^\mu) = (u, v, x, y)$$

Null Tetrad

Covariant components

$$l = (1) du$$

$$n = (H(u, x, y)) du + (1) dv + (W_x(u, x, y)^2) dx + (-W_y(u, x, y)^2) dy$$

$$m = \left(\frac{1}{2}\sqrt{2}\right) dx + \left(\frac{1}{2}i\sqrt{2}\right) dy$$

$$\bar{m} = \left(\frac{1}{2}\sqrt{2}\right) dx + \left(-\frac{1}{2}i\sqrt{2}\right) dy$$

Contravariant components

$$l = (-1) dv$$

$$n = (-1) du + (H(u, x, y)) dv$$

$$m = \left(-\frac{1}{2} \sqrt{2} W_x(u, x, y)^2 + \frac{1}{2} i \sqrt{2} W_y(u, x, y)^2 \right) dv + \left(\frac{1}{2} \sqrt{2} \right) dx + \left(\frac{1}{2} i \sqrt{2} \right) dy$$

$$\bar{m} = \left(-\frac{1}{2} \sqrt{2} W_x(u, x, y)^2 - \frac{1}{2} i \sqrt{2} W_y(u, x, y)^2 \right) dv + \left(\frac{1}{2} \sqrt{2} \right) dx + \left(-\frac{1}{2} i \sqrt{2} \right) dy$$

Directional Derivatives

$$D = -\partial_v$$

$$\Delta = -\partial_u + H(u, x, y) \partial_v$$

$$\delta = -\frac{1}{2} \sqrt{2} W_x(u, x, y)^2 + \frac{1}{2} i \sqrt{2} W_y(u, x, y)^2 \partial_v + \frac{1}{2} \sqrt{2} \partial_x + \frac{1}{2} i \sqrt{2} \partial_y$$

$$\bar{\delta} = -\frac{1}{2} \sqrt{2} W_x(u, x, y)^2 - \frac{1}{2} i \sqrt{2} W_y(u, x, y)^2 \partial_v + \frac{1}{2} \sqrt{2} \partial_x - \frac{1}{2} i \sqrt{2} \partial_y$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = i W_x(u, x, y) \frac{\partial}{\partial y} W_x(u, x, y) + i W_y(u, x, y) \frac{\partial}{\partial x} W_y(u, x, y)$$

$$\lambda = 0$$

$$\nu = -\sqrt{2}W_x(u, x, y) \frac{\partial}{\partial u} W_x(u, x, y) - i\sqrt{2}W_y(u, x, y) \frac{\partial}{\partial u} W_y(u, x, y) + \frac{1}{2}\sqrt{2} \frac{\partial}{\partial x} H(u, x, y) - \frac{1}{2}i\sqrt{2} \frac{\partial}{\partial y} H(u, x, y)$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = \frac{1}{2}i W_x(u, x, y) \frac{\partial}{\partial y} W_x(u, x, y) + \frac{1}{2}i W_y(u, x, y) \frac{\partial}{\partial x} W_y(u, x, y)$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = 0$$

$$\begin{aligned} \Phi_{12} = & \frac{1}{4}i\sqrt{2}W_x(u, x, y) \frac{\partial^2}{\partial x \partial y} W_x(u, x, y) + \frac{1}{4}i\sqrt{2} \frac{\partial}{\partial x} W_x(u, x, y) \frac{\partial}{\partial y} W_x(u, x, y) - \frac{1}{4}\sqrt{2} \frac{\partial}{\partial y} W_x(u, x, y)^2 - \frac{1}{4}\sqrt{2}W_x(u, x, y) \frac{\partial^2}{(\partial y)^2} W_x(u, x, y) \\ & + \frac{1}{4}i\sqrt{2} \frac{\partial}{\partial x} W_y(u, x, y)^2 + \frac{1}{4}i\sqrt{2}W_y(u, x, y) \frac{\partial^2}{(\partial x)^2} W_y(u, x, y) - \frac{1}{4}\sqrt{2}W_y(u, x, y) \frac{\partial^2}{\partial x \partial y} W_y(u, x, y) - \frac{1}{4}\sqrt{2} \frac{\partial}{\partial x} W_y(u, x, y) \frac{\partial}{\partial y} W_y(u, x, y) \end{aligned}$$

$$\Phi_{20} = 0$$

$$\begin{aligned}\Phi_{21} = & -\frac{1}{4}i\sqrt{2}W_X(u, x, y)\frac{\partial^2}{\partial x\partial y}W_X(u, x, y) - \frac{1}{4}i\sqrt{2}\frac{\partial}{\partial x}W_X(u, x, y)\frac{\partial}{\partial y}W_X(u, x, y) - \frac{1}{4}\sqrt{2}\frac{\partial}{\partial y}W_X(u, x, y)^2 - \frac{1}{4}\sqrt{2}W_X(u, x, y)\frac{\partial^2}{(\partial y)^2}W_X(u, x, y) \\ & - \frac{1}{4}i\sqrt{2}\frac{\partial}{\partial x}W_Y(u, x, y)^2 - \frac{1}{4}i\sqrt{2}W_Y(u, x, y)\frac{\partial^2}{(\partial x)^2}W_Y(u, x, y) - \frac{1}{4}\sqrt{2}W_Y(u, x, y)\frac{\partial^2}{\partial x\partial y}W_Y(u, x, y) - \frac{1}{4}\sqrt{2}\frac{\partial}{\partial x}W_Y(u, x, y)\frac{\partial}{\partial y}W_Y(u, x, y)\end{aligned}$$

$$\begin{aligned}\Phi_{22} = & W_X(u, x, y)^2\frac{\partial}{\partial y}W_X(u, x, y)^2 + 2W_X(u, x, y)W_Y(u, x, y)\frac{\partial}{\partial y}W_X(u, x, y)\frac{\partial}{\partial x}W_Y(u, x, y) + W_Y(u, x, y)^2\frac{\partial}{\partial x}W_Y(u, x, y)^2 - W_X(u, x, y)\frac{\partial^2}{\partial u\partial x}W_X(u, x, y) \\ & - \frac{\partial}{\partial u}W_X(u, x, y)\frac{\partial}{\partial x}W_X(u, x, y) + W_Y(u, x, y)\frac{\partial^2}{\partial u\partial y}W_Y(u, x, y) + \frac{\partial}{\partial u}W_Y(u, x, y)\frac{\partial}{\partial y}W_Y(u, x, y) + \frac{1}{2}\frac{\partial^2}{(\partial x)^2}H(u, x, y) + \frac{1}{2}\frac{\partial^2}{(\partial y)^2}H(u, x, y)\end{aligned}$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = 0$$

$$\begin{aligned}\Psi_3 = & \frac{1}{4}i\sqrt{2}W_X(u, x, y)\frac{\partial^2}{\partial x\partial y}W_X(u, x, y) + \frac{1}{4}i\sqrt{2}\frac{\partial}{\partial x}W_X(u, x, y)\frac{\partial}{\partial y}W_X(u, x, y) + \frac{1}{4}\sqrt{2}\frac{\partial}{\partial y}W_X(u, x, y)^2 + \frac{1}{4}\sqrt{2}W_X(u, x, y)\frac{\partial^2}{(\partial y)^2}W_X(u, x, y) \\ & + \frac{1}{4}i\sqrt{2}\frac{\partial}{\partial x}W_Y(u, x, y)^2 + \frac{1}{4}i\sqrt{2}W_Y(u, x, y)\frac{\partial^2}{(\partial x)^2}W_Y(u, x, y) + \frac{1}{4}\sqrt{2}W_Y(u, x, y)\frac{\partial^2}{\partial x\partial y}W_Y(u, x, y) + \frac{1}{4}\sqrt{2}\frac{\partial}{\partial x}W_Y(u, x, y)\frac{\partial}{\partial y}W_Y(u, x, y)\end{aligned}$$

$$\begin{aligned}\Psi_4 = & -W_X(u, x, y)\frac{\partial^2}{\partial u\partial x}W_X(u, x, y) + iW_X(u, x, y)\frac{\partial^2}{\partial u\partial y}W_X(u, x, y) - \frac{\partial}{\partial u}W_X(u, x, y)\frac{\partial}{\partial x}W_X(u, x, y) + i\frac{\partial}{\partial u}W_X(u, x, y)\frac{\partial}{\partial y}W_X(u, x, y) - iW_Y(u, x, y)\frac{\partial^2}{\partial u\partial x}W_Y(u, x, y) \\ & - W_Y(u, x, y)\frac{\partial^2}{\partial u\partial y}W_Y(u, x, y) - i\frac{\partial}{\partial u}W_Y(u, x, y)\frac{\partial}{\partial x}W_Y(u, x, y) - \frac{\partial}{\partial u}W_Y(u, x, y)\frac{\partial}{\partial y}W_Y(u, x, y) + \frac{1}{2}\frac{\partial^2}{(\partial x)^2}H(u, x, y) - i\frac{\partial^2}{\partial x\partial y}H(u, x, y) - \frac{1}{2}\frac{\partial^2}{(\partial y)^2}H(u, x, y)\end{aligned}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "*III*"